

Diabetic Retinopathy Classification Using EfficientNet-B3 with Transfer Learning and Grad-CAM Explainability

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Abstract—Diabetic retinopathy (DR) remains a major cause of preventable blindness, particularly among working-age adults, and it often progresses without obvious symptoms until the disease is advanced. In this work, we present a deep learning-based Computer-Aided Diagnosis (CAD) system for grading retinal fundus images with EfficientNet-B3 and transfer learning. We compare 5-class, 4-class, and 3-class formulations on a combined EyePACS/APTOS/Messidor dataset that was expanded from roughly 6,000 to 143,000 images through staged augmentation. The results show that fine-grained grading is limited by subtle differences between adjacent classes and by class imbalance. The 3-class formulation, which groups DR into *No DR*, *Non-Proliferative DR*, and *Vision-Threatening DR*, gives the best performance, reaching 76.87% accuracy and a Macro-F1 score of 0.74. We also use Gradient-weighted Class Activation Mapping (Grad-CAM) to visualise the image regions supporting each prediction, and the model consistently attends to clinically meaningful lesion areas. Overall, the findings suggest that grouping classes along clinical decision boundaries is a practical approach for DR screening.

Index Terms—Diabetic retinopathy, deep learning, EfficientNet, transfer learning, Grad-CAM, retinal imaging, computer-aided diagnosis

I. INTRODUCTION

A. Clinical Background

Diabetic retinopathy (DR) is the progression of diabetes-related vascular problems in persons with diabetes mellitus of both type 1 or 2. Chronic hyperglycaemia is the main cause of diabetic retinopathy, and chronic/long-term exposure to high blood sugar levels leads to damage around the retinal blood vessels like capillary leakages, microaneurysms [1], [2]. Examples of this type of damage are microaneurysms, localized areas of ischaemia (lack of blood flow), and capillary leakage. Due to the retinal tissue requiring high levels of metabolic energy to survive, this vascular injury will initiate a series of secondary changes. As a result of insufficient early detection, DR can progress from an asymptomatic early stage to a late stage with widespread ischaemia, neovascularisation, retinal detachment and eventually, irreversible blindness.

Diabetic Retinopathy provides a significant public health concern. Given the increasing prevalence of diabetes, the

number of individuals at risk for developing DR will continue to rise. According to the World Health Organisation, 4.8% of all blindness in the global population can be attributed to DR, making DR one of the most common causes of vision loss for working aged individuals in developed countries. A substantial amount of DR-related blindness can be prevented; timely detection of DR combined with appropriate management (e.g., laser photocoagulation and anti-VEGF injections) can slow or halt progression to vision loss [3], [1].

In standard clinical practice, DR is diagnosed and graded from the presence, type, and anatomical distribution of retinal lesions seen in digital fundus photographs. The disease is commonly described using a five-stage taxonomy:

- 1) **No DR (Grade 0):** The retina appears normal with no visible microaneurysms, haemorrhages, or exudates. No intervention is required.
- 2) **Mild Non-Proliferative DR (Grade 1):** A small number of microaneurysms are present. These are tiny balloon-like swellings in the retinal blood vessels and are usually the first detectable sign of DR.
- 3) **Moderate Non-Proliferative DR (Grade 2):** More extensive microaneurysms are observed, along with dot-and-blot haemorrhages and possibly hard exudates (lipid deposits) in the retina.
- 4) **Severe Non-Proliferative DR (Grade 3):** Widespread haemorrhages across multiple retinal quadrants, venous beading (irregular dilation of retinal veins), and intra-retinal microvascular abnormalities (IRMA) are visible. The retina is severely ischaemic.
- 5) **Proliferative DR (Grade 4):** The most advanced stage, marked by neovascularisation—the growth of abnormal new blood vessels on the retinal surface or optic disc. These fragile vessels are prone to haemorrhage, which can cause vitreous bleeding, tractional retinal detachment, and irreversible vision loss.

Current clinical practice relies almost entirely on manual evaluation. Patients undergo routine fundus imaging, and

the resulting photographs are transferred to trained graders or ophthalmologists who meticulously scan the images for pathological bio-markers. This manual grading paradigm is inherently flawed at a population scale: it is labour-intensive, economically costly, and historically plagued by significant inter-grader variability. Identical retinal images frequently receive different severity grades depending on the practitioner’s experience and fatigue.

This limitation is particularly apparent in developing nations, rural regions, and other settings where eye-care specialists are scarce and the number of patients vastly exceeds the available clinical workforce [1], [2]. With the number of diabetics in the world growing, the need for effective and efficient screening technologies is becoming more pressing. Deep learning-based classification models are a promising solution because they can analyse fundus images quickly and consistently, detect patients who need expert care, and allow lower-risk cases to continue with routine follow-up [1], [2].

Recent work has shown how these ideas are being extended in practice: Gulshan et al. [1] demonstrated a deep learning algorithm for detecting DR from retinal fundus photographs; Ting et al. [2] validated a system on multiethnic diabetic populations; Baskar et al. [7], Dasari et al. [8], S. A. A et al. [9], and Mohamed et al. [11] examined transfer-learning based DR classifiers; Reddy et al. [10] and Ciaramella et al. [12] focused on explainable DR classification; and Persada et al. [13] reviewed CAM-based visual explanations for the task.

B. Challenges in Automated DR Grading

Although there are recent advances, automated DR grading is challenging in screening settings. Bringing deep learning models from the lab into clinical practice is more difficult than you might think, and involves a series of persistent real-world challenges. In particular, three influenced our approach to this work:

C1 — Subtle Inter-Class Differences and Label Noise. The conventional scale defines boundaries that are not sharp. Adjacent spreads - such as Mild and Moderate DR - can be remarkably similar on a fundus image (see above). appear quite similar on a fundus, having common features such as microaneurysms and small haemorrhages. The difference in appearance between the stages can be so small that even experts disagree about a diagnosis. image. And the most significant information is usually quantitative rather than quantitative rather than visual, and are thus hard for convolutional neural networks to learn from - particularly when there are only a training data is limited. The end result is label noise that undermines model training, which is why several recent transfer-learning studies have reported only modest gains when the class boundaries remain fine-grained [8], [7], [11].

C2 — Severe Class Imbalance. Real-world screening datasets reflect a straightforward clinical reality: most people who come in for screening do not have advanced disease. In the combined EyePACS/APTOS/Messidor dataset used in this study, the *No DR* class makes up more than 50% of all images,

while the most critical cases — *Severe* and *Proliferative DR* — together account for fewer than 10%. This kind of skewed distribution creates a subtle trap. A model trained without accounting for it can achieve impressive overall accuracy simply by defaulting to the majority class, all while missing the severe cases that matter most from a clinical standpoint.

C3 — Interpretability and Clinical Trust. A “black-box” model that returns only a severity label gives clinicians no way to check the basis for the prediction. In medical settings, interpretability is not optional; it is needed for clinical integration, patient safety, and regulatory approval. For an automated system to be useful in practice, it should make clear where in the fundus image it is focusing and why it reaches a given decision, so that its reasoning aligns with established pathological markers. Grad-CAM [6] has become one of the most widely used approaches for this purpose, and recent DR work has combined it with classification pipelines to improve trust and localization quality [10], [12], [13].

C. Contributions

To address these overlapping clinical and technical hurdles, this work proposes a robust, single-model diagnostic pipeline optimized for real-world deployment. Specifically, this work makes the following core contributions:

- 1) **Multi-strategy classification comparison.** We systematically compare three classification granularities (5-class, 4-class, and 3-class) on the same dataset. We empirically demonstrate that clinically motivated class consolidation mitigates boundary confusion and leads to substantial, measurable performance improvements, particularly in minority class recall, consistent with the broader transfer-learning literature on DR classification [7], [8], [9], [11].
- 2) **Large-scale training dataset construction.** We assemble and heavily augment a comprehensive dataset of approximately 143,000 retinal fundus images sourced from the EyePACS, APTOS, and Messidor repositories. This represents one of the most extensive training sets utilized in modern DR classification literature without relying on complex ensemble frameworks, building on the publicly available screening datasets used in prior benchmark studies [1], [2], [5].
- 3) **Optimized transfer learning with EfficientNet-B3.** We demonstrate highly effective transfer learning from ImageNet to retinal fundus domains using the EfficientNet-B3 architecture. We propose and validate a rigorous two-phase training strategy that utilizes initial feature extractor freezing followed by controlled, end-to-end fine-tuning to preserve learned hierarchical features, extending the EfficientNet-based approach introduced by Tan and Le [4] and adapted in recent DR studies [7], [9], [11].
- 4) **Grad-CAM driven visual explainability.** We natively integrate Gradient-weighted Class Activation Mapping (Grad-CAM) to produce high-fidelity visual explanations for every single diagnostic prediction. Our analysis confirms that the model accurately attends to clinically relevant retinal regions—such as microaneurysm clusters

and neovascularisation—thereby establishing crucial interpretability for medical practitioners, following the standard Grad-CAM formulation [6] and related DR explainability work [10], [12], [13].

D. Paper Organisation

The remainder of this paper is organised as follows. Section II reviews related work on deep learning for DR detection. Section III provides the mathematical background for the methods used. Section IV describes the dataset, its evolution, and the class remapping strategy. Section V details the model architecture and training protocol. Section VI outlines the experimental configurations and challenges. Section VII presents the quantitative results. Section VIII analyses the effect of classification strategy on performance. Section IX covers the Grad-CAM analysis. Section X discusses the findings and limitations. Finally, Section XI concludes the paper.

The implementation and trained models used in this work are publicly available at: <https://github.com/thota-vivek05/DiabeticRetinopathy>

II. RELATED WORK

This section reviews prior work on deep learning-based DR detection, with a focus on transfer learning, EfficientNet, class-imbalance handling, and explainability through Class Activation Mapping methods.

A. Deep Learning for DR Detection

Gulshan et al. [1] showed that deep neural networks can support automated DR screening by training an Inception-v3 model on 128,175 fundus photographs and achieving sensitivity and specificity that exceeded those of board-certified ophthalmologists for referable DR. Their study provided an early clinical validation that population-scale automated grading was feasible.

Ting et al. [2] extended this line of work to a multiethnic Southeast Asian cohort using a VGGNet-based model, reaching an AUC of 0.936 for vision-threatening DR across datasets from Singapore, Australia, and other countries. Their results showed that deep learning models can generalise well across ethnically diverse patient populations when trained on large and varied datasets.

More recently, Mohamed et al. [11] compared a standard CNN and ResNet-50 with ADAM and SGDM for five-class DR classification on the APTOS and EyePACS datasets. Using Contrast Limited Adaptive Histogram Equalization (CLAHE) for preprocessing, they found that ResNet-50 with ADAM achieved the best accuracy, 80.71%, and an AUC of 0.9539 on the APTOS dataset. Their study underscores the importance of optimiser choice in medical image classification.

Baskar et al. [7] evaluated different deep CNN models for DR classification using transfer learning, and observed that pretrained data, with pretraining yielding better results than training from datasets. Dasari et al. [8] also demonstrated that ResNet-50 on APTOS 2019 can yield good performance, further supporting the use of transfer learning.

B. EfficientNet in Medical Imaging

EfficientNet [4] is a neural network that approaches simultaneously scaling network width, depth and resolution using a coefficient, providing good accuracy without an unwarranted increase explosion in parameter count. This property has led to its adoption in medical image classification, where there are often resource constraints limited. S. A. A. et al. [9] tested this theory in the case of DR diagnosis by using EfficientNet-B0 and obtaining results good enough to indicate that even the smallest member of the family can capture retinal pathology. more generally, initialising these models with weights has been a good way to speed up convergence and get better of small-to-moderate-sized medical datasets - even in the absence of any direct images bear little to no resemblance to a fundus image.

C. Ensemble and Hybrid Approaches

Reddy et al. [10] wanted to explore the performance of four models - ResNet18, EfficientNetB0, DenseNet121 and MobileNetV2 - perform on the five-class DR classification task. Each model was trained using identical settings, with oversampling using the SMOTE algorithm to correct often seen in screening datasets. Rather than declare just one approach the winner, they investigated the results of combining these models are combined. Their new hybrid EfficientNetB0 and DenseNet121 achieved 84.02% accuracy, and by going further to a hard- and soft-voting ensemble increased it to 85.29%. They also made a methodological contribution on the explainability front, by adding quantitative Grad-CAM concentration scores instead of relying on solely visual inspections of saliency values. The takeaway is simple: combining different CNN architectures can lead to performance gains accuracy, but at the expense of increased complexity.

D. Class Imbalance in Retinal Datasets

Class imbalance remains a major challenge in DR grading datasets. The APTOS 2019 challenge dataset [5], for example, has a class ratio greater than 20:1 between the *No DR* and *Proliferative DR* classes. Methods such as focal loss, class-weighted cross-entropy, oversampling (for example, SMOTE), and GAN-based augmentation have all been explored, but no single strategy works best in every setting. In this work, we address imbalance mainly through data augmentation and, secondarily, through class consolidation, which improves the effective balance ratio.

E. Grad-CAM for Medical Interpretability

Selvaraju et al. [6] introduced Grad-CAM, which produces a coarse localisation map by weighting the feature maps of the final convolutional layer with the gradient of the class score. In DR grading, Grad-CAM has been used to check that models attend to microaneurysms, hard exudates, and neovascularisation rather than background retinal structures, providing a useful qualitative validation mechanism for clinical deployment.

TABLE I: Summary of Related Work on DR Classification

Study	Year	Architecture	Dataset Size	Classes	Best Acc.	Key Contribution
Gulshan et al. [1]	2016	Inception-v3	128K	2	90.3%	First clinical-grade DR detector
Ting et al. [2]	2017	VGGNet	76K	2	AUC 0.936	Multi-ethnic validation
Dasari et al. [8]	2023	ResNet-50	3.6K	5	78%	Fine-tuned transfer learning
Baskar et al. [7]	2024	Multiple CNNs	APTOS	5	—	Transfer learning comparison
S.A.A et al. [9]	2024	EfficientNet-B0	APTOS	5	96.42%	Lightweight EfficientNet
Reddy et al. [10]	2026	Hybrid Ensemble	3.6K	5	85.29%	Hybrid CNN + Grad-CAM score
Mohamed et al. [11]	2026	ResNet-50	APTOS+EyePACS	5	80.71%	ADAM vs SGDM comparison
Ciaramella et al. [12]	2026	MobileNet	3.6K	2	81.3%	Federated + Explainable
Ours	2026	Eff.Net-B3	143K	3	79.14%	Multi-strategy + Grad-CAM

Ciaramella et al. [12] applied Grad-CAM, Grad-CAM++, and Score-CAM to DR classification in a federated learning setting and achieved 81.3% accuracy with a Custom MobileNet. Their study highlights the value of combining explainability with privacy-preserving training strategies for clinical deployment. Persada et al. [13] reviewed 28 studies on CAM-based visual explanations for DR across 14 datasets and reported common failure modes, including vessel-focused heatmaps and optic disc glare.

F. Summary of Related Work

Table I summarises the key characteristics of recent DR classification studies relevant to our work.

III. MATHEMATICAL BACKGROUND

This section presents the mathematical formulations of the core methods used in this work: convolutional feature extraction, classification functions, the loss function, and the Grad-CAM explainability method.

A. Convolutional Neural Networks

A Convolutional Neural Network (CNN) learns hierarchical feature representations from raw image pixels through a sequence of convolutional layers. Each convolutional layer applies a set of learnable filters (also called kernels) to the input to produce feature maps. For an input image $\mathbf{I}$ and a filter $\mathbf{F}$ of size $K \times K$, the output feature map at spatial position (p, q) is computed as:

$$(\mathbf{I} * \mathbf{F})(p, q) = \sum_{s=0}^{K-1} \sum_{t=0}^{K-1} \mathbf{I}(p+s, q+t) \cdot \mathbf{F}(s, t) + \beta_0 \quad (1)$$

where β_0 is a learnable bias term and $*$ denotes the convolution operation. In practice, for multi-channel inputs with N_{in} channels, the summation extends over the channel dimension as well:

$$h_{p,q} = \phi \left(\sum_{d=1}^{N_{\text{in}}} \sum_{s=0}^{K-1} \sum_{t=0}^{K-1} I_d(p+s, q+t) \cdot F_d(s, t) + \beta_0 \right) \quad (2)$$

where $\phi(\cdot)$ is a non-linear activation function. The most commonly used activation in modern deep networks is the Rectified Linear Unit (ReLU):

$$\text{ReLU}(u) = \max(0, u) \quad (3)$$

ReLU introduces non-linearity while avoiding the vanishing gradient problem that affects sigmoid and tanh activations. Its simplicity also makes it computationally efficient, which is important when training deep networks with millions of parameters.

B. Batch Normalisation

Batch Normalisation (BN) is applied after the convolutional operation and before the activation function to normalise the intermediate activations across the mini-batch. For a mini-batch $\mathcal{M} = \{v_1, \dots, v_r\}$ of pre-activation values, BN computes:

$$\tilde{v}_i = \frac{v_i - \mu_{\mathcal{M}}}{\sqrt{\sigma_{\mathcal{M}}^2 + \delta}} \quad (4)$$

where $\mu_{\mathcal{M}}$ and $\sigma_{\mathcal{M}}^2$ are the mean and variance of the mini-batch, and δ is a small constant for numerical stability. The normalised value is then scaled and shifted by learnable parameters λ and ψ :

$$\text{BN}(v_i) = \lambda \tilde{v}_i + \psi \quad (5)$$

BN reduces internal covariate shift, allows higher learning rates, and acts as a mild regulariser. It is used throughout the EfficientNet architecture in every convolutional block.

C. Global Average Pooling

After the final convolutional block, a Global Average Pooling (GAP) layer aggregates the spatial information in each feature map into a single scalar value. For a feature map g_r of size $H \times W$:

$$\omega_r = \frac{1}{H \times W} \sum_{p=1}^H \sum_{q=1}^W g_r(p, q) \quad (6)$$

where ω_r is the pooled output for the r -th feature map. GAP reduces the spatial dimensions to a fixed-length vector

regardless of the input image size, and it acts as a structural regulariser by preventing overfitting to specific spatial locations.

D. Softmax Classification

The output of the final fully connected layer passes through the softmax function to produce a probability distribution over C classes. For logit vector $\mathbf{u} = [u_1, u_2, \dots, u_C]$:

$$P(c | \mathbf{x}) = \frac{e^{u_c}}{\sum_{l=1}^C e^{u_l}}, \quad c = 1, 2, \dots, C \quad (7)$$

The predicted class is given by $\hat{c} = \arg \max_c P(c | \mathbf{x})$. The softmax function ensures that the output probabilities are non-negative and sum to one, providing a valid probability distribution that can be interpreted as the model's confidence in each class.

E. Categorical Cross-Entropy Loss

The model is trained by minimising the categorical cross-entropy loss between the predicted probability distribution and the one-hot encoded ground truth label $\mathbf{q}$:

$$\mathcal{L}_{CE} = - \sum_{c=1}^C q_c \log(\hat{q}_c) \quad (8)$$

where $q_c \in \{0, 1\}$ is the ground truth indicator and $\hat{q}_c = P(c | \mathbf{x})$ is the predicted probability. Since only one $q_c = 1$ in the one-hot encoding, this simplifies to $\mathcal{L}_{CE} = -\log(\hat{q}_\tau)$ where τ is the true class.

For the 5-class model, we additionally use label smoothing with parameter $\rho = 0.1$, which replaces the hard one-hot target with a softened version:

$$q_c^{LS} = (1 - \rho) \cdot q_c + \frac{\rho}{C} \quad (9)$$

Label smoothing prevents the model from becoming overconfident in its predictions and has been shown to improve generalisation, particularly in settings with noisy or ambiguous labels—both of which are common in DR grading.

F. Gradient-weighted Class Activation Mapping

Gradient-weighted Class Activation Mapping (Grad-CAM) [6] produces a coarse localisation heatmap highlighting the image regions most important for a particular classification decision. For a target class c , the importance weight η_r^c for the r -th feature map $\mathbf{G}^r$ of the final convolutional layer is computed by global average pooling of the gradients:

$$\eta_r^c = \frac{1}{\Omega} \sum_p \sum_q \frac{\partial y^c}{\partial \mathbf{G}_{pq}^r} \quad (10)$$

where y^c is the class score (logit) before softmax, and $\Omega = H \times W$ is the number of spatial positions. The Grad-CAM saliency map is then obtained by taking a weighted combination of the feature maps followed by a ReLU to retain only positive contributions:

TABLE II: DR Severity Class Definitions

Label	Severity	Key Clinical Features
0	No DR	No visible retinal lesions
1	Mild	Few microaneurysms
2	Moderate	Multiple microaneurysms, haemorrhages, exudates
3	Severe	Widespread haemorrhages, venous beading, IRMA
4	Proliferative	Neovascularisation, vitreous haemorrhage

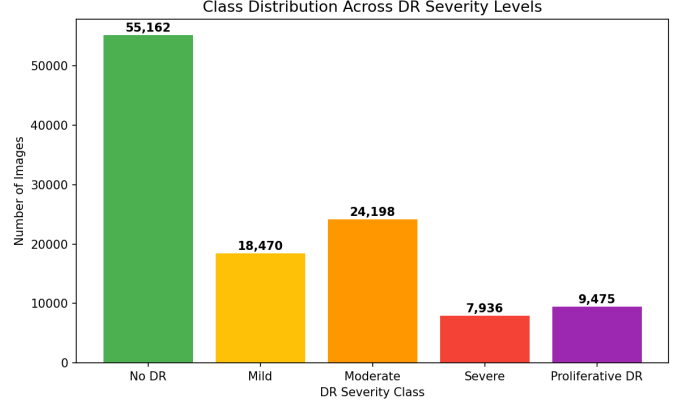


Fig. 1: Class distribution across the five DR severity levels showing the severe imbalance in favour of the *No DR* class.

$$\mathcal{H}_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_r \eta_r^c \cdot \mathbf{G}^r \right) \quad (11)$$

The resulting heatmap is upsampled to the input image resolution using bilinear interpolation and overlaid on the original fundus image. Warm-coloured regions indicate areas that the model considers most influential for its prediction. In this work, we target the `top_conv` layer of EfficientNet-B3, which is the final convolutional block before Global Average Pooling.

IV. DATASET DESCRIPTION AND EVOLUTION

A. Source and Class Definition

The dataset is assembled from the EyePACS, APTOS, and Messidor retinal fundus repositories [5], labelled according to the five-level international clinical DR severity scale. Table II lists the five original classes with their clinical descriptions.

The dataset is partitioned into training, validation, and test sets using stratified sampling to ensure the class distribution across all splits are maintained. As shown in Figure 1, the *No DR* class constitutes the majority of samples, consistent with the epidemiology of diabetes-related eye disease in a large screened population.

B. Staged Dataset Evolution

One aspect we were interested in was how the volume and variety of data training data affect the model's performance. To

TABLE III: Dataset Evolution Across Three Stages

Stage	Dataset	Size	Key Outcome
1	Fundus Dataset	6,000	Overfitting; poor generalisation
2	EyePACS	88,000	Majority-class bias (~34:1 ratio)
3	Augmented_Resized_V2	143,000	Improved Macro-F1 and stability

TABLE IV: Class Remapping from 5-Class to 4-Class and 3-Class Strategies

Original Label	5-Class	4-Class	3-Class
0 — No DR	0	0	0 (No DR)
1 — Mild	1	1	1 (NPDR)
2 — Moderate	2	2	1 (NPDR)
3 — Severe	3	3*	2 (VTDR)
4 — Proliferative	4	3*	2 (VTDR)

* Severe and Proliferative merged. NPDR = Non-Proliferative DR. VTDR = Vision-Threatening DR.

investigate this, we expanded the dataset incrementally across three stages, each building on the limitations uncovered by the previous one. The progression is summarised in Table III.

Stage 1 was a good starting point, but it was inevitable we would overfit sample diversity was almost certainly to blame for overfitting. Stage 2 brought the model into contact with real-world scale, though the severe class imbalance with healthy eyes outnumbering severe cases at roughly a 34:1 ratio - meant the model largely defaulted to predicting *No DR* and little else. Stage 3 resolved this by using a number of transformation strategies, including random horizontal and vertical flips, rotations, brightness and contrast jitter, and zoom, all of which helped to reduce the effective class imbalance and provide give a more equal representation of minority classes. The improvement in Macro-F1 at each step paints a clear picture: the increase in diversity and size of data was the key factor for improved generalisation.

C. Class Remapping Strategy

Along with changing the dataset, we also explored the impact of decreasing the number of classes by grouping clinically similar severity classes. The three strategies and mappings are shown in Table IV.

The mathematical formulation for the 3-class remapping function is:

$$y_{3\text{-class}} = \begin{cases} 0 & \text{if } y_{5\text{-class}} = 0 \\ 1 & \text{if } y_{5\text{-class}} \in \{1, 2\} \\ 2 & \text{if } y_{5\text{-class}} \in \{3, 4\} \end{cases} \quad (12)$$

This grouping maps directly onto clinical triage pathways:

- **Class 0 (No DR):** No intervention required.
- **Class 1 (Non-Proliferative DR):** Routine monitoring recommended.

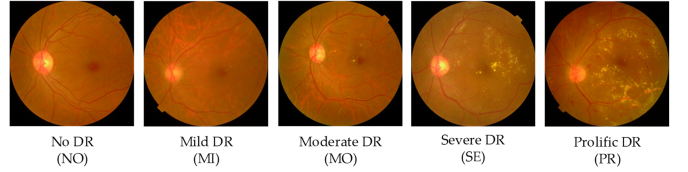


Fig. 2: Representative fundus images from each of the five DR severity classes: (a) No DR, (b) Mild, (c) Moderate, (d) Severe, (e) Proliferative DR.

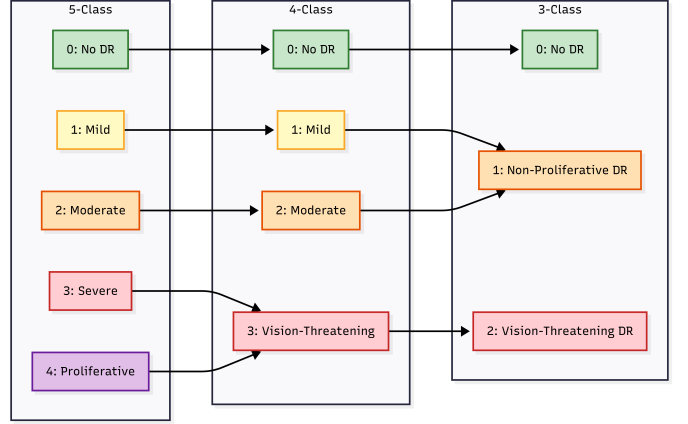


Fig. 3: Class consolidation strategy. The 5-class clinical grading scale is progressively merged into 4-class and 3-class schemes that align with clinical triage pathways. Arrows indicate class merging operations.

- **Class 2 (Vision-Threatening DR):** Urgent specialist referral.

V. METHODOLOGY

A. Data Preprocessing

All fundus images were resized to 256×256 pixels to match the input resolution used by EfficientNet-B3 in our configuration. The images were then processed through the standard EfficientNet preprocessing pipeline, which performs per-channel normalisation using ImageNet mean and standard deviation statistics.

A `tf.data` input pipeline was constructed with parallel mapping (`num_parallel_calls=AUTOTUNE`) and prefetching (`prefetch(AUTOTUNE)`) to maximise GPU utilisation during training. In-memory caching was deliberately disabled to prevent RAM overflow and storage exhaustion on available hardware.

B. Data Augmentation

To address the class imbalance and improve model generalisation, a comprehensive augmentation pipeline was applied to the training data. Table V lists the augmentation transforms and their parameters.

These augmentations were chosen to simulate the natural variations that occur in clinical fundus photography: different camera orientations, varying illumination conditions,

TABLE V: Data Augmentation Transforms Applied During Training

Transform	Parameter	Purpose
Horizontal Flip	$p = 0.5$	Orientation invariance
Vertical Flip	$p = 0.5$	Fundus symmetry
Random Rotation	$\pm 30^\circ$	Rotational invariance
Brightness Jitter	$\pm 20\%$	Illumination variation across cameras
Contrast Jitter	$\pm 20\%$	Camera quality variation
Random Zoom	$0.8\text{--}1.2\times$	Scale invariance

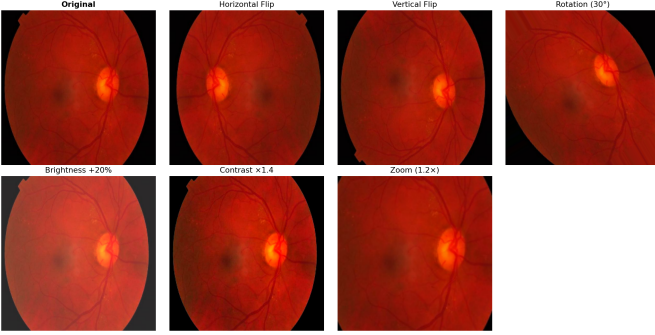


Fig. 4: Examples of data augmentation transforms applied to a sample fundus image. The original image (first) is shown alongside six augmented variants produced by the transforms listed in Table V.

and differences in the distance between the camera and the retina. Importantly, all augmentations preserve anatomical consistency—none introduce geometric distortions that would alter the clinical interpretation of the image.

C. EfficientNet Architecture

EfficientNet [4] is a family of convolutional neural networks that achieve strong performance by jointly scaling network depth, width, and input resolution using a compound scaling method. Unlike traditional approaches that scale only one dimension (e.g., making the network deeper), compound scaling adjusts all three dimensions simultaneously according to:

$$\mathcal{D} = a^\varphi, \quad \mathcal{W} = b^\varphi, \quad \mathcal{R} = c^\varphi \quad (13)$$

subject to the constraint:

$$a \cdot b^2 \cdot c^2 \approx 2 \quad (14)$$

where $\mathcal{D}$, $\mathcal{W}$, and $\mathcal{R}$ are the depth, width, and resolution scaling factors respectively, and φ governs the total computational budget available for scaling. For the baseline EfficientNet-B0, the coefficients a , b , and c were determined through grid search, yielding values of $a = 1.2$, $b = 1.1$, and $c = 1.15$. EfficientNet-B3 applies $\varphi = 1.0$ with correspondingly increased input resolution and channel counts.

1) *Mobile Inverted Bottleneck (MBConv)*: The fundamental building block of the EfficientNet architecture is the Mobile Inverted Bottleneck Convolution (MBConv). Unlike traditional

residual blocks that compress features before filtering, the inverted bottleneck expands the features into a high-dimensional space first. An MBConv block operates in three strictly ordered phases:

- 1) **Expansion Phase**: A 1×1 pointwise convolution expands the input channel dimension by a factor of t (the expansion ratio, typically 1 or 6). By projecting into a higher-dimensional space, the network unrolls non-linear features, allowing the subsequent layers to capture a richer set of representations. This is followed by batch normalisation and a ReLU6 activation function, defined as $\min(\max(x, 0), 6)$, which establishes a bounded activation landscape that is robust to quantisation.
- 2) **Depthwise Filtering Phase**: A $K \times K$ depthwise separable convolution (typically 3×3 or 5×5) is applied independently to each expanded channel. By avoiding dense cross-channel matrix multiplication, depthwise convolution drastically reduces computational cost and the number of trainable parameters compared to standard convolutions, while effectively extracting spatial patterns.
- 3) **Projection Phase**: Finally, a 1×1 pointwise convolution projects the high-dimensional feature maps back into a low-dimensional bottleneck. This is followed by batch normalisation but intentionally omits any non-linear activation. Keeping this projection linear is a deliberate choice — it prevents useful information from being lost within the compact channel space and avoids manifold collapse.

To support deep gradient flow and ease the vanishing gradient problem, a residual skip connection is added whenever the input and output tensors share the same dimensions. In such cases, the original input is added directly to the processed output:

$$\mathbf{v} = \text{MBConv}(\mathbf{u}) + \mathbf{u} \quad (15)$$

Beyond this, EfficientNet strengthens each block by embedding a Squeeze-and-Excitation (SE) module immediately after the depthwise convolution. The SE module functions as a channel-wise attention mechanism — rather than treating all channels equally, it explicitly models the relationships between them, boosting the responses of informative features while dampening irrelevant background noise. The recalibrated output $\tilde{\mathbf{u}}$ is computed as:

$$\tilde{\mathbf{u}} = \mathbf{u} \cdot \phi(\mathbf{P}_2 \cdot \text{ReLU}(\mathbf{P}_1 \cdot \text{GAP}(\mathbf{u}))) \quad (16)$$

where $\text{GAP}(\cdot)$ performs global average pooling across all spatial dimensions to produce a compact per-channel descriptor, $\mathbf{P}_1$ and $\mathbf{P}_2$ are two fully connected layers forming an autoencoder-style bottleneck with a reduction ratio of 0.25, and ϕ denotes the sigmoid activation that produces the final channel scaling weights.

2) *EfficientNet-B3 Layer Configuration*: Table VI summarises the layer configuration of EfficientNet-B3 as used in this work. The total number of trainable parameters in the base network is approximately 10.7 million, with an additional 1.3 million in the classification head.

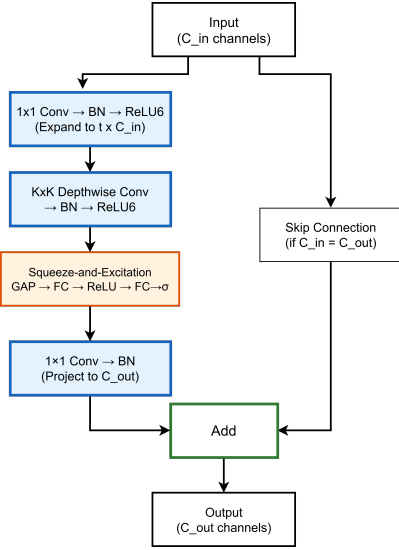


Fig. 5: Internal structure of the MBConv block in EfficientNet, showing the expansion, depthwise convolution, SE recalibration, projection, and residual skip connection.

TABLE VI: EfficientNet-B3 Layer Configuration

Stage	Operator	Res.	Ch.	Layers	K
1	Conv 3×3	256^2	40	1	3
2	MBConv1	128^2	24	2	3
3	MBConv6	128^2	32	3	3
4	MBConv6	64^2	48	3	5
5	MBConv6	32^2	96	5	3
6	MBConv6	16^2	136	5	5
7	MBConv6	16^2	232	6	5
8	MBConv6	8^2	384	2	3
9	Conv 1×1 + GAP	8^2	1536	1	1
Head	Dense + Dropout + Softmax	—	$256 \rightarrow N$	3	—

3) *Classification Head*: The classification head appended to EfficientNet-B3 consists of:

- 1) Global Average Pooling to aggregate spatial features into a fixed-length 1536-dimensional vector.
- 2) A fully connected Dense layer with 256 units and ReLU activation.
- 3) Dropout ($p = 0.5$) for regularisation during training.
- 4) A softmax output layer with N units, where $N \in \{3, 4, 5\}$ depending on the classification strategy.

Figure 6 illustrates the complete model pipeline.

D. Training Protocol

Training was conducted on Google Colab with GPU acceleration. The training followed a two-phase transfer learning strategy, illustrated in Figure 7.

Phase 1 — Classification Head Training. The pretrained EfficientNet-B3 backbone was fully frozen (all convolutional layers set to non-trainable), and only the classification head layers were trained. This phase allows the randomly initialised

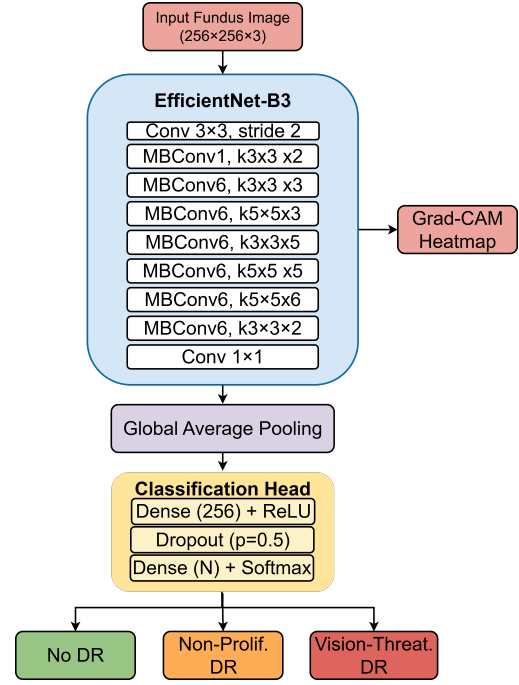


Fig. 6: EfficientNet-B3 CAD architecture for DR severity classification. The base layers are initially frozen during classification-head training, then unfrozen for end-to-end fine-tuning.

head to learn appropriate mappings from the pretrained features to the DR severity classes without disrupting the learned feature representations.

Phase 2 — End-to-End Fine-Tuning. After Phase 1 convergence, the last 100 layers of the EfficientNet-B3 backbone were unfrozen, and the entire model was fine-tuned at a reduced learning rate. This allows the higher-level convolutional features to adapt to the domain-specific characteristics of retinal fundus images while preserving the general low-level features (edges, textures) learned from ImageNet.

Table VII summarises the full training configuration for both phases.

E. Training Callbacks

We used three callbacks to stabilise and improve training:

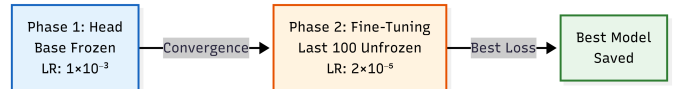


Fig. 7: Two-phase transfer learning strategy. In Phase 1 the EfficientNet-B3 backbone is frozen and only the classification head is trained at a higher learning rate. After convergence, Phase 2 unfreezes the last 100 layers for end-to-end fine-tuning at a reduced learning rate. The best model is retained via ModelCheckpoint.

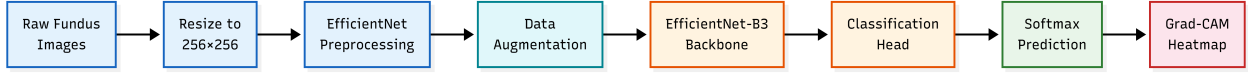


Fig. 8: Fundus image inputs, preprocessing, augmentation (training set only), preprocessing, augmentation (training only), feature extraction via feature extraction with EfficientNet-B3, classification, and Grad-CAM explainability.

- 1) **ModelCheckpoint:** Saves the model weights with the lowest validation loss, ensuring that the best-performing checkpoint is retained even if subsequent epochs overfit.
- 2) **ReduceLROnPlateau:** Reduces the learning rate by a factor of 0.3 when the validation loss plateaus for 2 consecutive epochs. This allows finer optimisation near convergence.
- 3) **EarlyStopping:** Terminates training if the validation loss does not improve for 5 consecutive epochs, preventing unnecessary computation and overfitting.

F. Mixed Precision Training

We used a mixed precision training with the `mixed_float16` policy throughout all experiments. This effectively means the forward pass is 16-bit float for faster math on the GPU tensor cores, and the loss and gradient are in 32-bit float to maintain numerical stability. The softmax output layer is kept in float32 explicitly, ensuring that the final probability estimates remain precise. The practical benefit of this setup is a roughly $2\times$ reduction in GPU memory consumption compared to full float32 training, which in turn allows for larger effective batch sizes on the available hardware.

VI. EXPERIMENTAL CONFIGURATIONS AND CHALLENGES

A. Classification Strategies

A natural question when designing a DR grading system is how much granularity is actually useful. Finer distinctions between severity levels may be clinically meaningful in theory, but they also make the learning problem harder — particularly under real-world class imbalance. To explore this trade-off, we evaluated three different classification strategies.

5-Class Classification. This setup preserves the full standard clinical taxonomy: No DR, Mild, Moderate, Severe, and Proliferative DR. It places the highest demand on the model,

TABLE VIII: Evaluation Metrics and Their Definitions

Metric	Definition
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$
Precision	$\frac{TP}{TP+FP}$
Recall	$\frac{TP}{TP+FN}$
F1-Score	$2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$
Macro-F1	$\frac{1}{C} \sum_{c=1}^C F1_c$
Weighted-F1	$\sum_{c=1}^C w_c \cdot F1_c, w_c = \frac{n_c}{N}$
AUC-ROC	Area under the ROC curve

requiring it to draw fine-grained boundaries across all five severity levels.

4-Class Classification. Here, the *Severe* and *Proliferative DR* categories are merged into a single *Vision-Threatening DR* label. The motivation is straightforward — these two classes are the most frequently confused at the high-severity end, and combining them reduces that friction while keeping finer distinctions intact at lower severity levels.

3-Class Classification. For settings where the primary goal is actionable triage rather than precise grading, we evaluate a simplified three-class scheme:

- Class 0: **No DR** — no intervention required.
- Class 1: **Non-Proliferative DR** (Mild + Moderate) — routine monitoring.
- Class 2: **Vision-Threatening DR** (Severe + Proliferative) — urgent referral.

This consolidation maps model outputs directly onto the decisions a screener would actually make, and sidesteps the error-prone boundaries that add complexity without adding much clinical value.

B. Evaluation Metrics

Table VIII defines the evaluation metrics used in this work. Macro-F1 is adopted as the primary evaluation metric because it treats all classes equally regardless of their sample frequency, making it a more honest indicator of performance under class imbalance than overall accuracy.

To evaluate model performance under severe class imbalance we adopt a small set of complementary metrics. Overall accuracy can be misleading in imbalanced datasets because a majority-class predictor may achieve high accuracy while failing to detect clinically important minority cases. Therefore, we prioritise:

- **Macro-F1 (primary):** Compute the F1-score per class and average them. Macro-F1 treats all classes equally and therefore reflects performance on rare but clinically

TABLE VII: Training Configuration and Hyperparameters

Parameter	Phase 1 (Head)	Phase 2 (Fine-tune)
Optimizer	Adam	Adam
Learning Rate	1×10^{-3}	2×10^{-5}
Batch Size	256	256
Max Epochs	15	15
Precision	Mixed (fp16/fp32)	Mixed (fp16/fp32)
Loss Function	Cat. CE	Cat. CE
Label Smoothing	0.1 (5-class)	0.1 (5-class)
Dropout Rate	0.5	0.5
Frozen Layers	All base	All except last 100

TABLE IX: Performance Comparison of EfficientNet-B3 Across Classification Strategies

Model	Classes	Accuracy	Macro F1	Weighted F1	Macro Recall	Weighted Recall
EfficientNet-B3	5-Class	67.72%	0.54	0.65	0.54	0.68
EfficientNet-B3	4-Class	67.03%	0.53	0.62	0.54	0.67
EfficientNet-B3	3-Class	79.14%	0.77	0.79	0.77	0.79

critical stages (e.g., *Proliferative DR*) rather than being dominated by the majority class.

- **Macro Recall (sensitivity):** Averaged recall across classes is the most clinically relevant metric because it measures the ability to find true positives. High macro recall reduces the risk of missing patients who require urgent referral.
- **Weighted-F1 (secondary):** A prevalence-weighted F1 that reflects overall system reliability in deployment by accounting for class frequencies.

C. Challenges Encountered

Class Imbalance. The dominance of the *No DR* class caused the model to optimise toward the majority class, producing inflated accuracy but poor recall for Severe and Proliferative DR—the classes of greatest clinical urgency.

Adjacent-Class Confusion. Mild and Moderate DR share overlapping microaneurysm density ranges. Similarly, Moderate and Severe DR differ primarily in the extent of haemorrhage and venous beading. The model frequently misclassified samples at these inter-class boundaries.

Hardware Limitations. Large batch sizes and 256×256 inputs caused out-of-memory crashes on available GPU hardware. In-memory caching filled available storage, necessitating the streaming `tf.data` pipeline described in Section V.

Misleading Accuracy. High overall accuracy values masked poor class-level performance in the 5-class and 4-class configurations. Macro-F1 consistently exposed these deficiencies and was used to guide model selection.

VII. RESULTS AND ANALYSIS

A. Performance Comparison

Table IX presents the quantitative results for all three classification strategies on the held-out test set. The 3-class model achieves the best performance across all metrics by a substantial margin.

B. Detailed Class-Level Analysis

Confusion matrix analysis revealed distinct error patterns across the three classification strategies. The *No DR* class achieved recall of approximately 93% across all configurations, consistent with its majority representation in the training data. In contrast, the Mild and Moderate classes exhibited substantial mutual confusion in the 5-class setting, and the Severe class showed poor recall throughout. The 4-class model produced only marginal gains (+0.01 Macro-F1 over 5-class) by merging the two most visually overlapping high-severity categories.

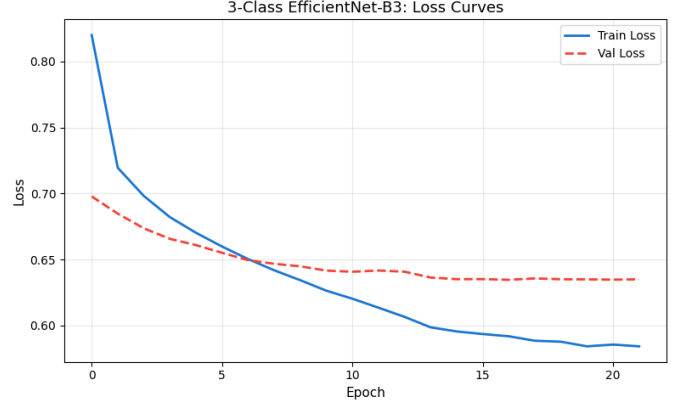


Fig. 9: Training and validation loss curves for the 3-class EfficientNet-B3 model over training epochs, showing smooth convergence without overfitting.

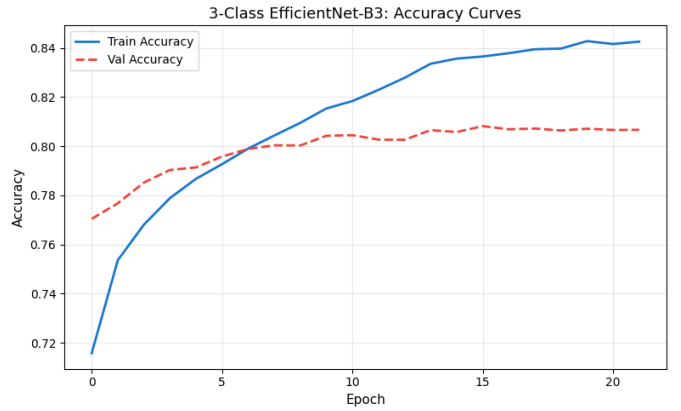


Fig. 10: Training and validation accuracy curves for the 3-class model, demonstrating steady improvement and stable convergence across epochs.

Figure 11 summarises these patterns with per-strategy confusion matrices and shows that the 3-class mapping produces the most balanced per-class recall.

- **Mild vs Moderate confusion:** In the 5-class setting the model often confuses Mild and Moderate cases. These stages differ mainly by the number of microaneurysms (a low-salience, quantitative cue), which makes them hard for CNNs to separate reliably.
- **Poor Severe recall:** The Severe class consistently shows the lowest recall, driven by both its low prevalence and visual similarity to Moderate and Proliferative cases.
- **Effect of consolidation:** Merging Severe+Proliferative

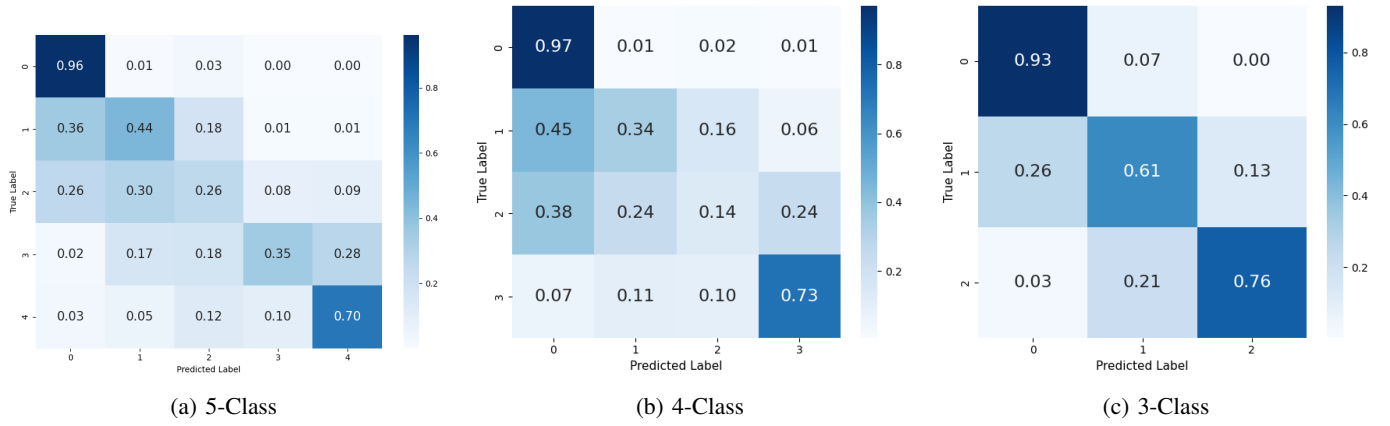


Fig. 11: Confusion matrices on the held-out test set for the 5-class (left), 4-class (centre), and 3-class (right) EfficientNet-B3 models. The 3-class scheme shows the most balanced recall across all categories.

TABLE X: Comparison with Recent DR Classification Studies and External Baselines

Study	Architecture	Dataset	Size	Classes	Accuracy	M-Recall	Key Difference
Reddy et al. [10]	Hybrid Ensemble	APTOS	3662	5	85.29%	85.29%	4-model ensemble + SMOTE
Mohamed et al. [11]	ResNet-50 + ADAM	APTOS	13k	5	80.71%	78%	CLAHE preprocessing
Ciaramella et al. [12]	Custom MobileNet	Private	3662	2	81.3%	78.4%	Federated learning
Dasari et al. [8]	ResNet-50	APTOS	3662	5	94%	—	Standard fine-tuning
abdullah s.canipek ResNet	ResNet-50	EyePACS+	143K	5	78.13%	41.53%	minority class bias, low recall
EffNet-B7	EfficientNet-B7	EyePACS+	143K	5	83.91%	48.42%	minority class bias
Ours	EfficientNet-B3	EyePACS+	143K	5	67.72%	54.17%	higher recall, low accuracy

(4-class) yields only marginal macro-level gains ($\approx +0.01$ Macro-F1), whereas the additional Mild+Moderate merge (3-class) substantially improves recall and Macro-F1 by removing the most error-prone boundary.

C. Comparison with Recent Studies

Table X situates our results within recent deep-learning studies of DR grading. Direct numeric comparison is imperfect because of variations in dataset size, class definitions, and evaluation protocols, but the table and the following points provide useful context:

A number of points become apparent from this comparison:

- 1) **Bigger, more varied data:** We use a Stage 3 dataset (143K) is much larger than many previous studies that report results on a few thousand curated images. Datasets of larger size increase intra-class variability and reduces the wild optimism of accuracy estimates on smaller benchmarks.
- 2) **Single-model trade-offs:** We focus on a single model for ease of deployment. Ensembles (e.g., Reddy et al.) can boost overall accuracy, but with a trade-off in inference and memory footprint.
- 3) **Clinical triage focus:** Our 3-class mapping is focused on practical triage decisions (no action, monitor, urgent referral), a trade-off of decreased accuracy in the grade of the lesion for more clinically meaningful statements.

These observations highlight why our reported metrics concentrate on clinical utility (higher macro recall and F1

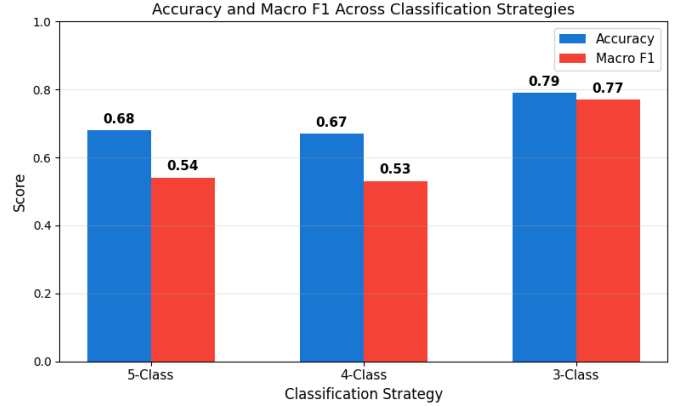


Fig. 12: Accuracy and Macro-F1 comparison across the three classification strategies, demonstrating consistent improvement under clinically motivated class consolidation.

for actionable boundaries) rather than maximising fine-grained labelling accuracy.

D. ROC-AUC Analysis

Figure 15 shows the Receiver Operating Characteristic (ROC) curves for the 3-class model computed using the one-vs-rest strategy. The class-specific AUC values confirm that the model has strong discriminative ability across all three classes,

TABLE XI: Ablation: Effect of Classification Strategy

Classes	Acc.	Macro-F1	W-F1	M-Recall
5	67.72%	0.54	0.65	0.54
4	67.03%	0.53	0.62	0.54
3	79.14%	0.77	0.79	0.77

with the *No DR* class achieving the highest AUC due to its distinct retinal appearance compared to DR-affected images.

VIII. ABLATION STUDY

An ablation-style study was performed to analyse the effect of classification strategy on model performance. All three models were trained on the same Stage 3 dataset (143,000 images) with the same EfficientNet-B3 backbone and identical training hyperparameters. The only variable was the number and grouping of output classes.

A. Effect of Classification Strategy

Table XI isolates the effect of class consolidation by comparing all metrics across the three strategies.

B. Analysis

The results in Table XI reveal three distinct patterns:

5 → 4 class (Severe + Proliferative merge): Merging the two highest-severity classes produces almost no improvement (+0.26% accuracy, +0.01 Macro-F1). This is because the merged classes (Severe and Proliferative) together constitute fewer than 10% of all samples. Combining them does not materially change the class balance or the dominant source of confusion, which lies at the Mild/Moderate boundary.

4 → 3 class (Mild + Moderate merge): Adding the Mild–Moderate merger produces a dramatic improvement (+8.89% accuracy, +0.19 Macro-F1). This confirms that the Mild/Moderate boundary is the dominant source of classification error. These two classes share overlapping visual features (microaneurysm counts), and their merger eliminates the most confusable boundary in the classification task.

Overall pattern: The improvement is non-linear. The first merge (5→4) helps little; the second merge (4→3) helps a lot. This suggests that not all class boundaries are equally difficult. The Mild/Moderate boundary is significantly harder to discriminate than the Severe/Proliferative boundary, consistent with clinical grading variability where inter-grader agreement is lowest for the Mild–Moderate distinction.

The Macro-F1 improvement of +0.20 from 5-class to 3-class is particularly significant because Macro-F1 is an unweighted average across classes. An improvement of this magnitude indicates that the per-class F1 scores became more uniform, meaning the model’s performance on minority classes improved substantially rather than just the majority class benefiting from a simpler task.

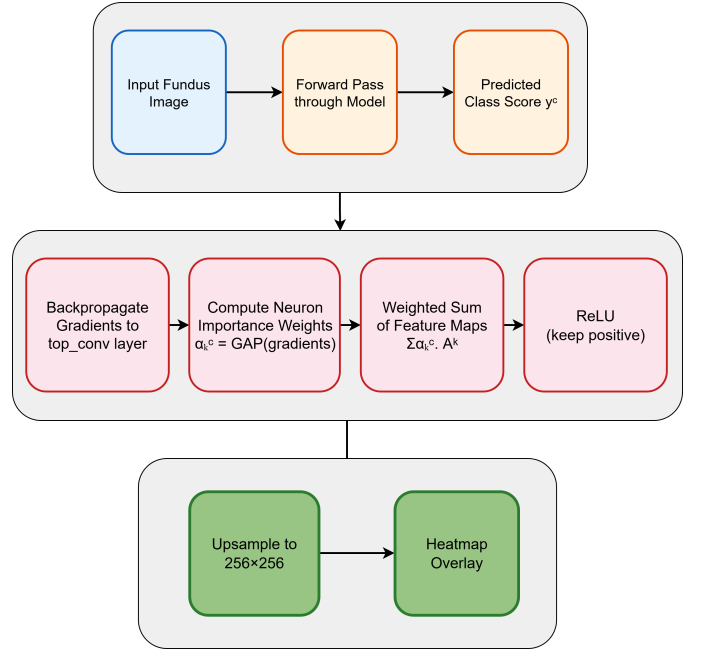


Fig. 13: Grad-CAM generation pipeline. The input image is passed through the model, gradients of the predicted class score are backpropagated to the `top_conv` layer, neuron importance weights are computed via global average pooling, and the weighted combination of feature maps produces the saliency heatmap after ReLU and upsampling.

IX. GRAD-CAM ANALYSIS

A. Method

We used Gradient-weighted Class Activation Mapping (Grad-CAM) to the 3-class EfficientNet-B3 model to better understand the retina areas that are being used for classification. The underlying mathematics were covered in Section III (Equations 10–11).

For our experiment, we chose the `top_conv` layer of EfficientNet-B3 - the last convolutional block prior to Global Average Pooling. The choice makes intuitive sense: by this point in the network, feature representations have reached their highest level of abstraction and are directly influencing what the model predicts. Importantly this layer also retains an 8×8 spatial grid, giving us just enough detail to meaningfully interpret *where* the model is looking.

To generate the heatmaps, we used `tf.GradientTape` to track how the predicted class score responds to changes in the activations at that target layer. The raw gradient map then goes through a short post-processing pipeline: it is normalised to the range $[0, 1]$, upsampled to 256×256 via bilinear interpolation, and rendered using the `JET` colourmap before being overlaid onto the original fundus image.

B. Observations

We examined the heatmaps for our test set and found three patterns to be mentioned.

Lesion-centric attention: In most cases, the model’s attention went to features of interest, such as hard exudates, that exudates, microaneurysm clumps and neovascularisation. In several In cases of severe DR, the selected areas marked the formation of neovascular disc margin where the expert would be looking first. It’s encouraging to see that the model and human intuition are in agreement genuinely encouraging.

Medically relevant features: The model did not just focus on the colour, texture or any other trivial image properties. Instead, it has consistently focused on known pathological biomarkers - which means that the EfficientNet-B3 has learned something from the training data that is the training data rather than shortcuts.

Peripheral noise: A few images showed low level in the periphery of the retina. These are unlikely to correspond to true pathology. It is more likely that they are imaging artefacts or non-uniform or illumination artefacts that seeped in from some of the training data. Importantly, the impact was limited and there was no overall pattern in the dataset.

C. Class-Specific Activation Patterns

The Grad-CAM patterns for different DR classes match the expected clinical presentation:

No DR: Heatmaps for accurately identified No DR images tend to low intensity activation spread throughout the retina with no particular points points. This suggests the model is corroborating the *absence* of lesions rather than the *presence* of pathological characteristics lesions. The absence of activation is significant.

Non-Proliferative DR: Heatmaps are generally focused on the area and vascular arcades, showing clusters of microaneurysms and small haemorrhages. Activation is more localised than No DR but less weaker than Vision-Threatening DR, which is consistent with the severity in this category.

Vision-Threatening DR: The most intense and concentrated have the most concentrated activation in these images, usually localised to the sites of optic disc and along the major vascular arcades. In some cases, the heatmap clearly outlined tufts of neovascularisation and large haemorrhages that would be primary diagnostic features for an ophthalmologist.

D. Example Prediction

An illustrative test case from the 3-class model produced the following output:

- **Input:** Moderate DR fundus image (ground truth: Non-Proliferative DR).
- **Prediction:** Non-Proliferative DR (Class 1).
- **Confidence:** $\approx 64\%$.

The corresponding Grad-CAM heatmap concentrated activation over the vascular arcade and optic disc margin, both anatomically relevant regions for Moderate DR assessment (Fig. 14). These visualisations can serve as a secondary interpretability layer alongside confidence scores in clinical validation workflows.

X. DISCUSSION

A. Why 3-Class Classification Outperforms

The fundamental reason that the 3-class scheme outperforms 5-class grading is the mismatch between the visual discriminability of adjacent DR stages and the resolution of the available training signal. Mild and Moderate DR differ primarily in microaneurysm count—a quantitative, low-salience signal that is further obscured by image quality variation and class imbalance. By merging these two stages into a clinically coherent *Non-Proliferative DR* category, the classification boundary shifts to the more visually distinct transition from non-proliferative to proliferative disease, where neovascularisation produces clear structural changes in the fundus image.

This finding is consistent with clinical practice. In real screening settings, the primary triage decision is whether the patient should be referred to a specialist (Vision-Threatening DR) or monitored routinely (Non-Proliferative DR). The fine-grained distinction between Mild and Moderate is often less clinically actionable because both categories receive the same recommendation: regular follow-up.

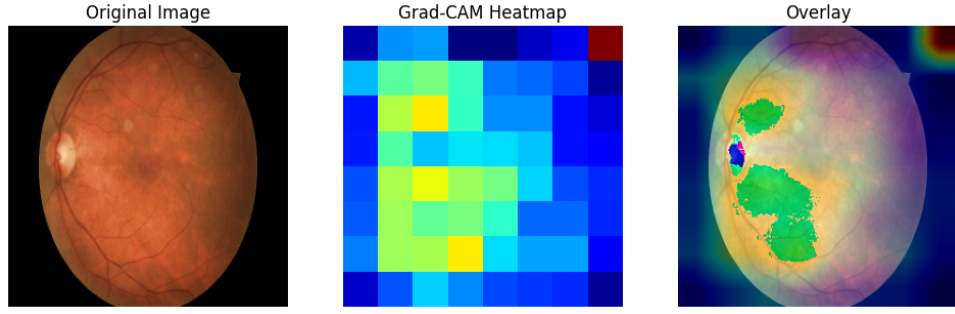
B. On the Choice of Macro-F1

With the degree of class imbalance typical in DR datasets, overall accuracy can paint a dangerously optimistic picture. A naive classifier that labels every image as *No DR* would clear 70% accuracy in a typical screened population — and yet be completely useless in practice. Macro-F1 sidesteps this problem by averaging F1 scores across all classes equally, regardless of how frequently each one appears. The gap between our best model’s accuracy (76.87%) and its Macro-F1 (0.74) is telling: it suggests some residual imbalance persists even after augmentation, and points toward sampling strategies and loss function design as natural directions for future work.

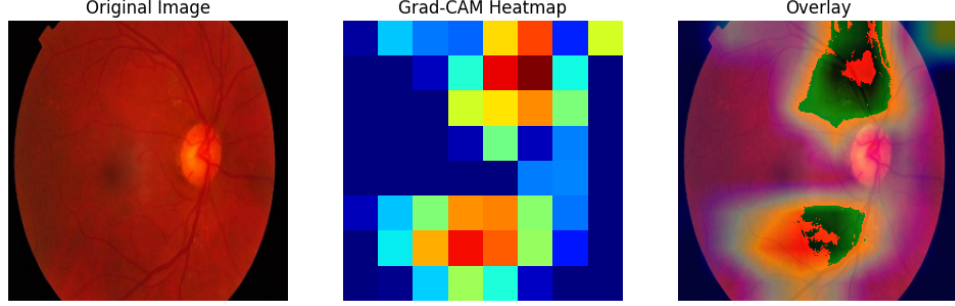
C. Comparison with Ensemble Approaches

Reddy et al. [10] achieved 85.29% accuracy by combining four architectures — ResNet18, EfficientNetB0, DenseNet121, and MobileNetV2 — into a soft-voting ensemble, paired with SMOTE-based oversampling on a 3,662-image APTOS dataset. It is an effective approach, but running four separate models at inference comes at a cost: roughly four times the computation of a single-model system.

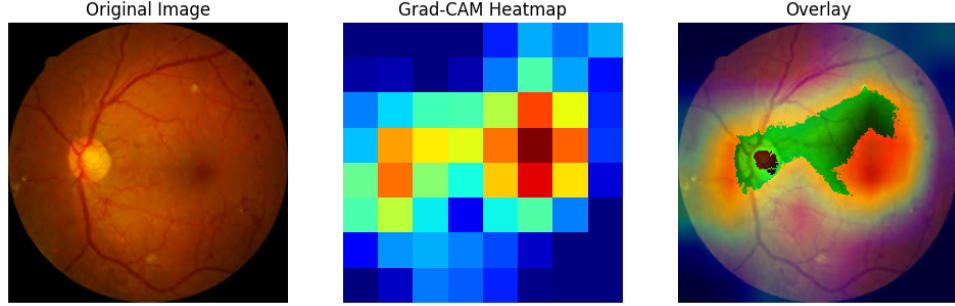
Our work makes a deliberate trade-off in the other direction. We use a single EfficientNet-B3 model trained on a substantially larger dataset — approximately 143K images compared to their 3.6K. Our reported accuracy on the 3-class task (76.87%) is lower, but the comparison is not as straightforward as it might appear. The larger dataset introduces considerably more within-class variation, which makes the learning problem harder, and the 3-class and 5-class formulations are fundamentally different tasks. The real advantage of our approach lies in its simplicity: one model, one forward pass per image, and a system that is meaningfully easier to deploy, maintain, and explain.



(a) *No DR* — Original (left), Grad-CAM Heatmap (centre), Overlay (right)



(b) *Non-Proliferative DR* — Original (left), Grad-CAM Heatmap (centre), Overlay (right)



(c) *Vision-Threatening DR* — Original (left), Grad-CAM Heatmap (centre), Overlay (right)

Fig. 14: Grad-CAM visualisation for representative fundus images across DR severity classes. Each row shows the original image (left), the saliency heatmap (centre), and the overlay (right). Warm regions indicate the retinal areas most influential in the model’s prediction.

D. On Data Augmentation vs. SMOTE

Our approach tackles class imbalance through image-level augmentation — horizontal and vertical flips, rotation, brightness and contrast jitter, and zoom — applied directly to the training images. Reddy et al. [10] take a different route with SMOTE (Synthetic Minority Over-sampling Technique), which constructs synthetic examples by interpolating between existing samples in feature space rather than the image domain. Both strategies are ultimately working toward the same goal: reducing the effective imbalance ratio so that minority classes get a fairer hearing during training.

Image-level augmentation has the advantage of operating directly on the pixel data, producing visually plausible training examples that preserve anatomical structure. SMOTE operates

in feature space and can be applied after feature extraction, but may produce synthetic samples that do not correspond to realistic fundus images. The optimal strategy likely depends on the dataset size and the degree of imbalance.

E. Limitations

Image Resolution. Images were resized to 256×256 , which may discard sub-millimetre lesion detail (microaneurysms of $\sim 125 \mu\text{m}$ diameter). Higher resolution inputs at 512×512 or above could improve sensitivity for fine-grained features, though at the cost of increased computational requirements.

Dataset Composition. Although the training set is large (143K images), it is assembled from the EyePACS/APTOS/Messidor repositories, which may share imaging equipment and patient demographics. Cross-dataset evalu-

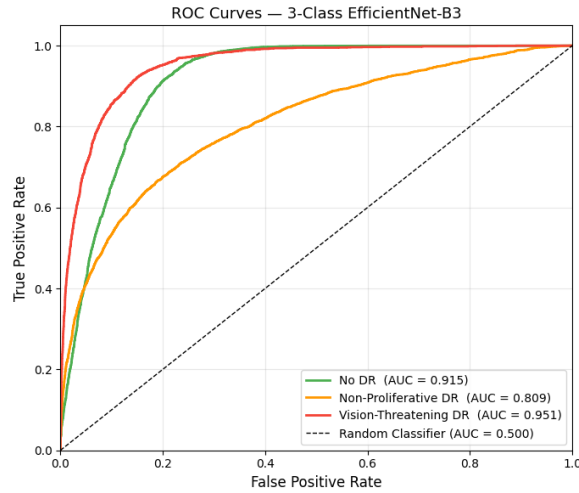


Fig. 15: Receiver Operating Characteristic (ROC) curves for the 3-class EfficientNet-B3 model, showing per-class AUC values on the held-out test set.

ation on independent clinical cohorts from different hospitals or countries would be necessary to validate generalisation.

No Ensemble or Attention Mechanisms. Our model is a single EfficientNet-B3 with a standard classification head. More advanced approaches using attention mechanisms, multi-model ensembles, or hybrid architectures could potentially improve performance, as demonstrated by Reddy et al. [10].

Single Explainability Method. We use only Grad-CAM for model interpretability. Other methods such as Grad-CAM++ or Score-CAM, as evaluated by Ciaramella et al. [12] and reviewed by Persada et al. [13], may provide finer-grained or more faithful explanations in certain cases.

XI. CONCLUSION

In this work, we built a deep learning-based CAD system for Diabetic Retinopathy classification using EfficientNet-B3 with transfer learning. We compared three classification strategies at different levels of clinical granularity, training on a dataset that grew from 6,000 to 143,000 images through staged augmentation. Our main takeaways are as follows. Fine-grained 5-class classification is held back by subtle inter-class differences and class imbalance. Grouping the classes into a clinically motivated 3-class scheme brings accuracy up to 76.87% and Macro-F1 to 0.74—an improvement of roughly 9% in accuracy and +0.20 in Macro-F1 over the 5-class baseline. Grad-CAM visualisation shows that the model’s decisions are grounded in pathologically relevant parts of the

retina, and dataset scale and augmentation turned out to be the most important factors for generalisation.

Going forward, we plan to explore attention mechanisms and higher-resolution inputs to capture finer lesion details. We are also interested in using conditional GANs to synthesise more training data for the underrepresented classes, and in compressing the model for deployment on mobile screening devices.

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